

REAL-TIME STUDY ROOM BOOKING APP

React Native + Expo SDK 57 | Short technical report

1. GENERAL INFORMATION AND DELIVERABLES

Student: Dương Bảo Đạt (23IT046), class 23SE1, Faculty of Computer Science, VKU. Contribution: 100%. Instructor: Nguyen Thanh Tuan, PhD.

Live demo: vku-study-room-booking.pages.dev | Repository: github.com/duongdatdev/vku-study-room-booking

APK: [GitHub Release v1.0.0](#). Direct-install Android preview build.

2. FEATURE CHECKLIST

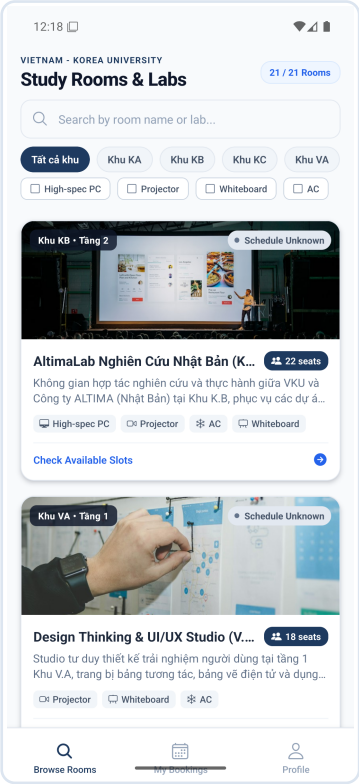
FEATURE	IMPLEMENTATION / OBSERVATION
Discovery and filters	21 rooms rendered; building KA filter shows 7. Search, capacity and equipment filters are in the app.
List performance	FlatList with memoized cards and bounded render windows. Frame rate was not measured.
Booking conflict engine	Seven campus-local dates and 2-hour slots. Database trigger and unique index reject collisions.
VKU authentication	Supabase email OTP restricted to @vku.udn.vn; email delivery currently fails before verification.
Booking pass	QR pass, server-confirmed check-in and cancellation implemented; live booking not yet demonstrated.
Cross-device sync	Owner-scoped bookings and sanitized occupancy via Supabase Realtime; second-device test pending.
Notifications	15-minute local reminder scheduled after server confirmation; delivery test pending.
Offline handling	Room browsing remains available; booking stops when shared schedule or network is unavailable.

3. ARCHITECTURE

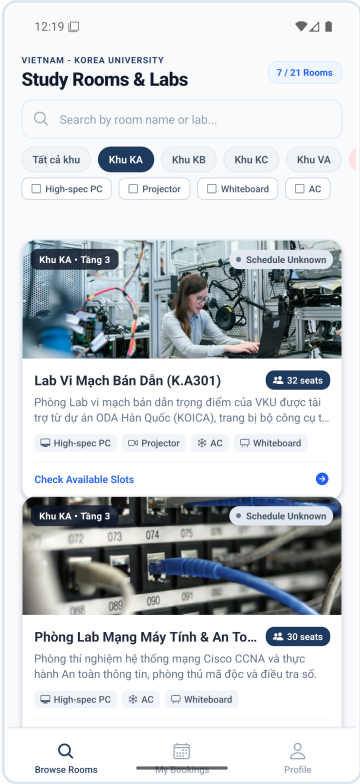
Expo UI and React Navigation → Zustand (filters and current snapshots) → Supabase Auth / Postgres / Realtime (authoritative reservations). Row-level security scopes private bookings to their owner. A separate occupancy table contains only room, date and slot identifiers.

4. ANDROID SCREENSHOTS

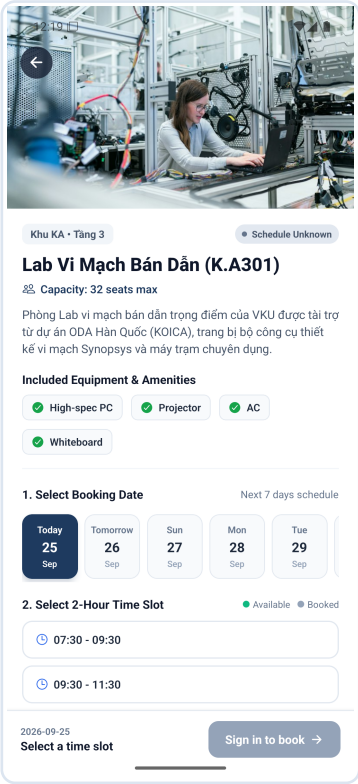
Pixel 6 emulator, API 36 | AVD stored at D:\Android\avd
Captured from the installed native Android release APK; images are not mockups.



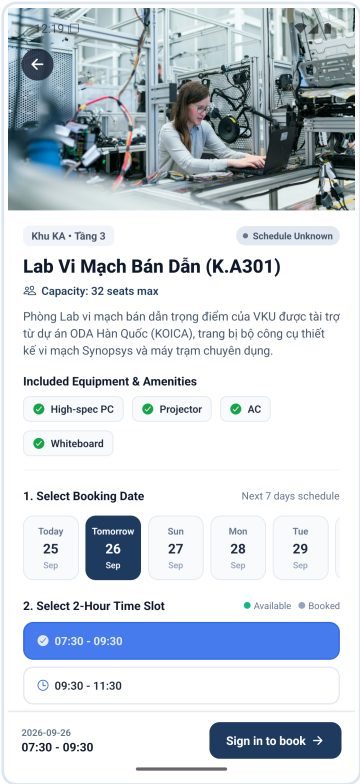
Room discovery: 21 of 21 rooms



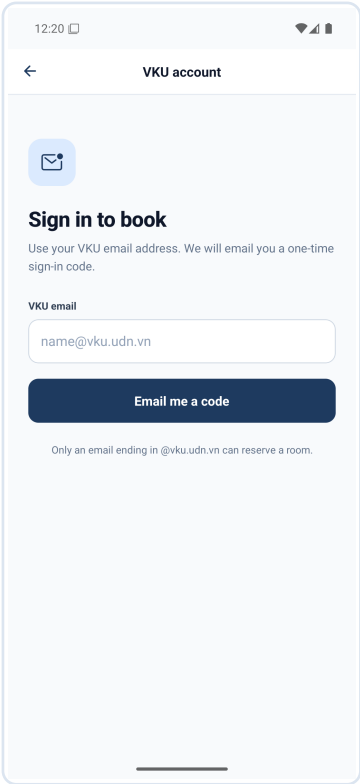
Khu KA filter: 7 of 21 rooms



Room detail and 7-day selector



Tomorrow, 07:30-09:30 selected



VKU email sign-in screen

5. ENGINEERING NOTES AND RESULTS

CONCURRENT BOOKING

The slot list is a usability snapshot. A Postgres partial unique index is the final guard against two active bookings for one room/date/slot; the insert response controls whether the app displays a pass. Database validation also checks the authenticated VKU email and seven-day campus window.

PRIVACY AND SYNCHRONIZATION

Students read and mutate only their own booking rows through row-level security. `room_occupancy` publishes room/date/slot keys without student details. Supabase Realtime triggers refreshes; Zustand stores the current view, not the authoritative record. Dates are evaluated in Asia/Ho_Chi_Minh on the client and in the database.

OBSERVED ANDROID BEHAVIOR

The native APK installed and opened on the Pixel 6 API 36 emulator. The room list displayed 21 rooms, the KA building chip reduced the list to 7, and the room detail screen allowed a 2-hour slot selection. Supabase REST reads for `rooms` and `room_occupancy` returned HTTP 200. The `Schedule Unknown` badge was captured between defined campus time slots and does not mean that all rooms were available.

OPEN VERIFICATION ITEM

A live OTP request to the supplied VKU address returned Error sending confirmation email from Supabase. Custom SMTP must be configured before a new user can authenticate. The authenticated booking, QR pass, cancellation, second-device update and reminder delivery were therefore not captured or claimed as end-to-end verified. Expo Go also cannot load this Android notification module on SDK 57; use the native APK.

REPRODUCE AND COMPLETE THE DEMONSTRATION

1. Apply `supabase/schema.sql` and `supabase/seed.sql`; set the Supabase URL and publishable key as documented in README. 2. Configure verified-domain SMTP and set the Confirm signup and Magic Link templates to include `{{ .Token }}`. 3. Install the GitHub APK, sign in with a VKU inbox, book a free slot, and capture the QR pass. 4. Sign in on a second device to observe owner-scoped reservations and shared occupancy.